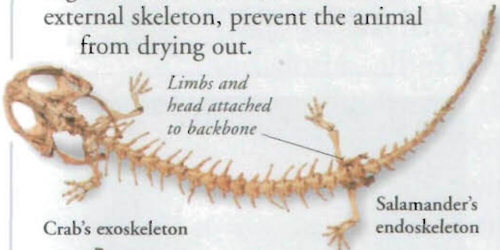
Animal skeletons

The skeleton is a supportive framework that maintains the shape of an animal and enables it to move. Most skeletons are hard structures, either inside or outside the animal's body, to which muscles are attached. The skeleton may also protect internal organs and, in the case of an insect's external skeleton, prevent the animal from drying out.



Limbs and head attached to backbone

Salamander's endoskeleton

Crab's exoskeleton

Internal skeletons

A skeleton found inside the body is called an endoskeleton. Most vertebrates have a skeleton made of cartilage and bone. Joints between the bones allow the animal to move. The endoskeleton grows with the rest of the body.

External skeletons

A hard outer skeleton that covers all or part of the body is called an exoskeleton. An insect's outer cuticle and a snail's shell are examples of an exoskeleton. An insect's exoskeleton does not grow and must be shed, or moulted, periodically to allow the animal to grow.

Hydrostatic skeleton

The hydrostatic skeleton is an internal skeleton found in soft-bodied animals such as earthworms. It consists of a fluid-filled core surrounded by muscles, and maintains the shape of the worm.

Earthworm

Worm gets longer when it contracts its muscles.

Movement of an eel through water

Eel moves by throwing its body into curves that push against the water.

Moving in water

Many aquatic animals are adapted for movement in water by having streamlined bodies. Most fish move by pushing their tail fin from side to side. This pushes the water backward and sideways, and propels the fish forward. Whales move in a similar way, except that the tail moves up and down.

Animal movement

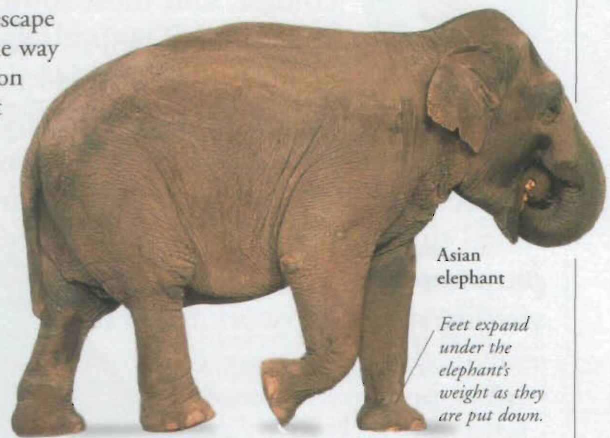
The ability to move is characteristic of animals, which move to find food, escape from predators, and find a mate. The way in which an animal moves depends on its complexity, lifestyle, and where it lives. The wide range of movement includes swimming through water, walking and creeping on land, and flying or gliding in air.

Wings sweep downward to produce forward thrust.

Young chaffinch in flight

Movement in air

Insects, birds, and bats are capable of powered flight using wings. Birds have lightweight, streamlined bodies. They use energy to flap their wings, which pushes them forward. As air passes over the wings it creates the lift that keeps the bird in the air.



Asian elephant

Feet expand under the elephant's weight as they are put down.

Movement on land

Animals move on land in a variety of ways. Many have limbs that raise the body off the ground, support it, and enable the animal to walk, run, or hop. The animals move forward by pushing the ends of their legs, or feet, backward against the ground.

Animal senses

The main senses are vision, hearing, taste, smell, and touch. Animals use their senses to find out what is going on around them. A stimulus from outside, such as a sound, is detected by a sense organ, such as the ear. Nerve impulses from sense organs are interpreted by the animal's brain which "decides" how to respond.

Dragonfly eyes

Eyes

Eyes contain sensors that are sensitive to light. When stimulated they send nerve impulses to the brain, which enable it to build up a picture. Insects have compound eyes made up of many separate units, or ommatidia.

Longhorn beetle

Antennae

These are found on the head of arthropods such as insects. They are used to detect odours and may detect chemicals called pheromones released by insects to communicate with each other. Antennae also detect vibrations and movements in the air or in water.

External ear flaps channel sounds into the ear.

Domestic Basenji dog

Ears

Some animals can detect sounds with ears. The ear converts sounds into nerve impulses that can be interpreted by the animal's brain. Animals use sounds to communicate with each other and to detect approaching predators or prey.

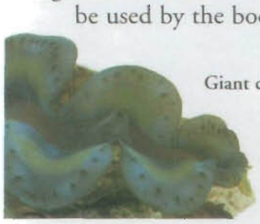
Feeding

All animals feed by taking in food. They use a range of feeding strategies and can be grouped accordingly. Some animals kill and eat others, some graze or browse on plants, others filter food particles from water. After feeding, or ingestion, food is digested so that it can be used by the body.

Mormon caterpillar consuming a leaf

Herbivores

Animals that feed solely on plants are called herbivores. Many use specialized mouthparts, such as grinding teeth, to break up tough plant tissues. Plant material is not a rich food source, and most herbivores eat a lot to obtain the necessary nutrients.



Giant clam

Filter feeders

These are animals that feed by sieving food particles from water that flows into their body. Many are sedentary and draw in a current of water. Some whales are filter feeders that eat small animals called krill.

Dragonfly larva with stickleback

Carnivores

These types of feeders are adapted to detect prey animals, to catch and kill them, and to cut them up to eat them. They include cats, eagles, and some insects. Dragonfly larvae live in water and they can catch small fish to eat.

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